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Tourism, Space, and Environmental Change: Exploring the Socio-Spatial Transformation of Karangsong, Indonesia

Idrus Jamalulel[1], Li-Wei Liu[2*]

[1] Department of Business Administration, College of Management, Chaoyang University of Technology, 413310 Taichung City, Taiwan
[2] Department of Leisure Services Management, College of Management, Chaoyang University of Technology, 413310 Taichung City, Taiwan

* Correspondence: Li-Wei Liu (llouis@cyut.edu.tw)

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Abstract: This study examines the socio-spatial transformation associated with coastal tourism in Karangsong, Indramayu, Indonesia, focusing on community livelihoods, place meanings, and mangrove conservation. An interpretive qualitative case study combined semi-structured interviews with 47 participants from six stakeholder groups, non-participant observation in September 2025, and analysis of regional planning documents. Data were analyzed using reflexive thematic analysis informed by Lefebvre's Production of Space. Five interconnected themes concern coastal uses, livelihood diversification, place identity, tourism–conservation relations, and stakeholder priorities. Observations documented tourism facilities, food businesses, guiding, and visitor activities alongside existing livelihood activities and mangrove conservation areas. Participants associated tourism with supplementary economic opportunities and changing meanings of the coast, while expressing concerns about uneven benefits and environmental pressures. The regional spatial plan identifies Karangsong as a tourism location and links mangrove protection with education and tourism objectives. Together, these materials support an interpretation of coastal transformation through the layering and reconfiguration of uses and meanings, as tourism becomes an additional function within a landscape that retains livelihood, conservation, and community roles. The case illustrates how everyday practices, institutional representations, and lived meanings intersect in the production of coastal space. Participants' recollections, current observations, and planning objectives are distinguished, keeping the interpretation bounded to the Karangsong case and the available evidence.

Keywords: Coastal tourism; Socio-spatial transformation; Community livelihoods; Mangrove conservation; Production of space; Karangsong

1. Introduction

Coastal tourism has become an important form of tourism development because coastal environments combine natural attractions, cultural resources, economic activities, and opportunities for recreation. Increasingly, beaches, mangrove ecosystems, fishing areas, waterfronts and coastal settlements are being incorporated into tourism development, creating new economic opportunities but simultaneously placing pressure on environmentally sensitive landscapes and local livelihoods. Recent geographical academic work has highlighted the interdependence between space and place and tourism, and has called for more attention to the role of tourism in the processes of place-making, spatial relations and integration and future development of places (Córdoba Azcárate, 2025). In this sense, destinations are continuously performed through the interaction of economic interests, government planning, community practice, tourist activity and environmental processes (Ghulyan, 2024; Sun et al., 2020).

The idea of production of space suggested by Lefebvre (1991) provides a relevant theoretical framework for the analysis of these transformations. Lefebvre conceived space as socially produced through the interplay of spatial practices, representations of space and representational spaces. Tourism studies have been increasingly interested

in issues of space and place beyond the physical distribution of attractions to address issues of spatial justice, place-making, memory, community identity and unequal access to tourism spaces (Jamal & Higham, 2021).

According to Lapointe et al. (2021), from a socio-spatial viewpoint influenced by the works of Lefebvre, it was determined that development in coastal tourism can lead to significant changes in a coastal zone by making tourism branches of economy be a part of the processes of spatial adaptation. Recent studies from Indonesia found that the development of coastal tourism could be associated with economic opportunities for the residents but there are still some issues to be solved such as environmental degradation, infrastructure development, and institutional coordination (Tuanaya, 2026). Overall qualitative studies about coastal tourism in Indonesia show that community participation is critical to make the management of tourism work and ensure that economic processes are correlated with environmental sustainability (Rozaki et al., 2025).

This is a significant issue in the Indramayu Regency of West Java, Indonesia, where various coastal resources and developing tourism activities are present. A good example is the area called Karangsong with mangrove rehabilitation and coastal tourism flourishing together with the traditional professions of coastal communities. Previous research has established the ecological and tourism potential of Karangsong. Previous research examines coastal land-use changes, mangrove rehabilitation, and ecotourism potential in Karangsong (Susana et al., 2020). Other studies assess community-based tourism, biophysical suitability, and sustainable management, while identifying shortcomings in facilities and tourism management (Beddu et al., 2026; Prihadi et al., 2024).

Although these studies provide valuable knowledge concerning ecological suitability, carrying capacity, community-based tourism, tourism potential, and management strategies, they leave an important socio-spatial dimension insufficiently explored. Much of the existing research asks whether the coastal environment is suitable for tourism, how tourism can be managed, what infrastructure is required, or which strategies can improve destination sustainability. The gap is of theoretical significance as well. The lens of production-of-space has been used in previous tourism research on ethnic communities and urban or heritage destinations (Su & Sun, 2020), but less attention has been paid to the production of coastal tourism spaces in smaller Indonesian destinations, particularly from an interpretive qualitative perspective. Existing research on Karangsong has generated important ecological and managerial knowledge, but there remains limited understanding of how different actors interpret the incorporation of tourism into an existing livelihood and conservation landscape. Accordingly, this study conceptualizes coastal tourism development as a process of socio-spatial transformation rather than simply tourism expansion. This study explores the interaction between tourism development and existing livelihood practices, environmental conditions, spatial uses, place identities, and stakeholder interests in the coastal area of Karangsong, Indramayu. It addresses how these changes are experienced and interpreted by local residents, tourists, tourism actors, government stakeholders, and environmental actors, with particular attention to their potentially competing priorities concerning tourism development, local livelihoods, and environmental conservation.

Therefore, this study aims to investigate the socio-spatial transformation of coastal tourism in Karangsong, Indramayu, Indonesia, particularly in terms of evolving spatial practices, local livelihoods, meanings of place, perceptions of the environment, and stakeholder relations. This study is guided by four research questions, namely (1) How has tourism development changed the social and spatial features of coastal areas in Karangsong, Indramayu? (2) How do local communities perceive the changes in use and significance of coastal spaces related to tourism? (3) How do tourism development, local livelihoods, and environmental conservation relate to each other in these changing coastal spaces? and (4) What priorities do different stakeholders express regarding the use and development of coastal space in Karangsong?

By answering these questions, this study contributes in three ways. First, it contributes theoretically by applying the production of space perspective to coastal tourism development in an Indonesian context. Second, it contributes empirically by providing qualitative evidence concerning how tourism transforms a coastal destination beyond measurable indicators of tourism suitability or environmental carrying capacity. Third, it makes a practical contribution by showing how local meanings of place, livelihood interests, environmental concerns and stakeholder perspectives can be integrated into sustainable tourism planning.

2. Literature Review

2.1 Tourism, Space, and Place

Tourism is a spatial phenomenon. Tourism takes place between places, and is embedded in geographically, socially, culturally and environmentally specific contexts (Williams & Lew, 2014). But space is not a passive container of tourism, it is actively produced and reconfigured through mobility, investment, infrastructure, representation, consumption and everyday practices. Recent research in tourism geography stresses the co-constitutive relationship of tourism, space and place, and calls for more research attention to spatial justice, place-making, memory, identity and lived experience of tourism (Córdoba Azcárate, 2025). The distinction between space and place is particularly important for understanding destinations. Space can be defined as a relational field that is produced through social interaction, while place is given meaning through people's experiences, memories,

practices and attachments. Tourism can thus contribute to place-making by assigning new meanings and functions to existing spaces. Tomassini & Lamond (2023) suggest that tourism space is relational and is constantly constructed and influenced by different "power geometries". Their perspective directs attention to who produces, controls, uses, and experiences tourism space, and how its benefits and costs are distributed.

2.2 Production of Space and Tourism Development

One way to think about these relationships is to use the theory of production of space. According to Lefebvre (1991), space is not just a neutral physical container, but it is produced socially. His theory distinguishes between three dimensions of space: spatial practice, representation of space, and representational spaces. This model is increasingly used in tourism studies, as tourism sites are produced through interaction of the actors involved: governments, businesses, communities, tourists, and so on. Yu & Zhong (2022) also applied this framework in the study of waterfront tourism. Their study of Guangzhou's Pearl River waterfront illustrates how government actors, infrastructure development, and spatial planning contribute to the formation of tourism space. The results of their research indicate that tourism space is built only through negotiations between governmental bodies and tourism industry actors, rather than automatically resulting from the presence of natural or cultural attractions. More recently Yang et al. (2026) found that tourism intensification can incorporate residential spaces into tourism networks, while simultaneously encouraging residents to sustain traditional lived spaces through everyday practices. The authors' findings point to a tension between tourism-induced spatial commodification and community-based lived space.

2.3 Socio-Spatial Transformation and Tourism

Socio-spatial transformation concerns changes in the relationships between space and social life, including land use, migration, mobility, social relations, place identity, and meanings. Tourism can create economic opportunities while raising concerns about commercialization, displacement, social injustice, and environmental degradation. The spatial effects of tourism have been thoroughly analyzed under the framework of spatial justice. For example, Díaz-Parra & Jover (2021) show that in the development of tourism there are processes of place alienation where local people have less control of their everyday spaces. One of the studies found that the question of who benefits from socio-spatial transformation should be considered in the study of tourism development processes (Paddison & Hall, 2023).

2.4 Coastal Tourism Development in Indramayu

Coastal tourism is especially sensitive, as its development occurs in ecosystems that often have a high ecological value, while supporting local livelihoods at the same time. Mangroves, beaches, estuaries, fisheries and coastal wetlands provide environmental services, while at the same time acting as tourism attractions. Therefore, the development of coastal tourism involves a continuous relationship between economic use, environmental preservation and community livelihoods. According to Garsetiasih et al. (2021), it has been observed that a positive perception toward mangrove preservation is associated with economic rewards in Karangsong, with the local residents being instrumental in keeping the stakeholders motivated toward conservation practices. This means that development of tourism and conservation will enhance one another only if the natives feel the benefits and are actively engaged in the processes. Still, such a relationship is not always smooth. Similarly, the tourism growth can exert pressure on the ecosystem of the region and lead to conflicting objectives of the economic development versus environmental conservation. As highlighted by Novianti et al. (2022), in the process of conserving mangroves in Karangsong, significant tourism and socioeconomic benefits can be achieved; however, it is important to remember about the necessity of maintaining harmony between economic growth and environmental possibilities. Hence, the sustainability of coastal tourism should be viewed as a socio-environmental negotiation rather than an environmental management problem. It is particularly important to understand how different actors perceive and negotiate these conflicting priorities for destinations undergoing tourism transformation. The earlier studies conducted in Indramayu, including research done by Prihadi et al. (2024) demonstrated the potential of Karangsong regarding ecotourism development. To assess Karangsong, the authors used the combination of community-based tourism and biophysical indicators, as a result of which the authors determined the appropriateness of mangrove tourism in the area at 83.7%, as well as its carrying capacity of about 803 visitors per day. In a broader context, research on the development of eco-villages based on mangrove resources has been conducted by Anjani et al. (2023) in several coastal areas in Indramayu, namely Pabean Udik, Karangsong, Dadap, and Pabean Ilir. Their research findings showed that tourism, fisheries, micro, small, and medium-sized enterprises, and mangrove ecosystems are important factors for the regional development, and the concept of green mapping can be used for visualization of available resources as well as development opportunities. All of these studies taken together reveal the ecological and tourism potential of the coastal regions of Indramayu.

2.5 Conceptual Model: Production of Space

Although existing studies provide important knowledge about coastal tourism development in Indramayu, an important conceptual and empirical gap remains. First, previous research has predominantly examined the characteristics and suitability of coastal resources. Studies of Karangsong have assessed mangrove potential, tourism suitability, carrying capacity, and ecological conditions (Garsetiasih et al., 2021; Prihadi et al., 2024; Susana et al., 2020). These studies establish what resources exist and whether they are appropriate for tourism. Second, another stream of research has focused on management and development strategies, including community-based tourism, stakeholder involvement, SWOT analysis, and tourism planning (Anjani et al., 2023; Novianti et al., 2022). These studies provide important recommendations concerning how tourism should be developed. However, comparatively little attention has been given to what happens to the social and spatial character of coastal places as tourism develops. The present study therefore uses Lefebvre's spatial triad not simply to classify different aspects of coastal tourism, but to examine how multiple uses and meanings are produced within the same geographical space. The Karangsong case provides an opportunity to extend the application of the Production of Space perspective to a coastal setting in which tourism does not simply replace an existing livelihood landscape. Instead, fishing, aquaculture, tourism, mangrove conservation, community activities, and visitor recreation remain materially and socially co-present. This suggests that socio-spatial transformation may occur through the layering, overlap, and reconfiguration of existing functions rather than through a straightforward transition from one dominant use of space to another. The case also draws attention to the relational character of the spatial triad: everyday practices, institutional representations, and lived meanings are not produced independently, but interact as different actors seek to maintain, use, develop, protect, or experience the same coastal environment. In this sense, the empirical contribution is to show how a coastal tourism space can remain multifunctional while its relative meanings, uses, and priorities are being reconfigured through tourism development.

Figure 1 combines the conceptual orientation that guided the study with the final themes developed through the empirical analysis and the resulting planning implications. It was developed by Lefebvre (1991) and the existing literature on tourism, coastal transformation, livelihoods, environmental conservation, place identity, and stakeholder relations. The initial conceptual orientation was used as a sensitizing framework rather than as a predetermined set of empirical themes. In particular, the three dimensions of perceived space (spatial practices), conceived space (representations of space), and lived or representational spaces provided the initial analytical orientation for examining how coastal tourism may transform the social production of space. Perceived space refers primarily to the material and everyday use of coastal space, including patterns of movement, access, livelihood activities, tourism activities, business practices, and interactions among different uses. In the Indramayu case, fishing, aquaculture, tourism businesses, visitor movement, mangrove activities, and the use of tourism facilities may therefore be examined as spatial practices when the analytical focus is on what people actually do in and through coastal space. For example, the presence of tourism facilities is not itself sufficient to classify them as spatial practices; rather, their use by visitors, residents, or tourism operators constitutes the relevant spatial practice. Conceived space concerns how coastal space is planned, organized, regulated, designed, and institutionally represented. In this case, tourism plans, infrastructure development strategies, environmental management arrangements, destination branding, and government representations of tourism development are relevant to conceived space. Lived space concerns the meanings, experiences, memories, identities, and symbolic associations through which coastal space is experienced. In Indramayu, the coast may simultaneously be experienced as a livelihood environment, a tourism destination, a place of community identity, and an environmental landscape. For example, a resident's account of how the coast has changed may provide evidence of lived space when it expresses changing attachment, identity, memory, or meaning. Similarly, a visitor's experience of the mangrove area contributes to the analysis of lived space when the focus is on how the place is perceived and experienced rather than simply on the visitor's physical activity. The final themes were developed and refined through the subsequent reflexive thematic analysis.

The three dimensions therefore interact rather than operate sequentially or independently. This distinction also means that the analytical classification is determined by the aspect of the empirical material being examined, rather than by the object itself. Existing livelihoods are not automatically classified as spatial practices, tourism development is not automatically conceived space, and participant accounts are not automatically lived space. Fishing, for instance, is spatial practice when examined as an everyday livelihood activity, but participants' memories or meanings associated with fishing may contribute to lived space, while fisheries zoning or government regulation may be examined as conceived space. Similarly, tourism activities may represent spatial practice when examining actual visitor behavior, conceived space when examining their planning or institutional organization, and lived space when examining the meanings and experiences generated through those activities. The conceptual model treats socio-spatial transformation as emerging from the interaction among changing material practices, institutional representations, and lived meanings. This approach provides a more precise basis for interpreting how tourism, livelihoods, environmental conservation, and community identities become interconnected within the changing coastal landscape of Indramayu. The framework identifies several issues that were considered relevant

to the empirical investigation, including changing spatial uses, access and control, livelihood opportunities, environmental conservation, place meanings, stakeholder relations, and possible future directions of coastal development. These issues should not be interpreted as final themes generated from the data. Instead, they represent areas of inquiry derived from the theoretical framework and previous literature. The subsequent qualitative analysis used deductive and inductive coding to examine how these issues appeared in the empirical material and whether they formed coherent patterns across interviews, observations, and documents.

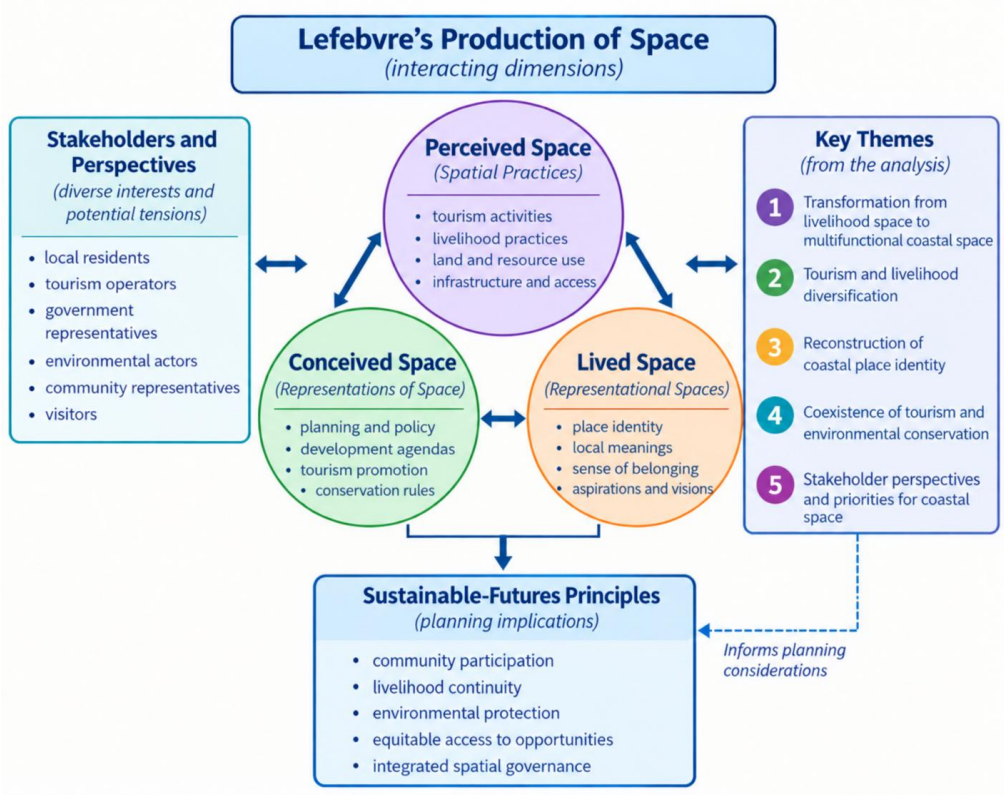


Figure 1. Framework, themes, and planning implications informed by Lefebvre (1991)

3. Methodology

This research project utilized a qualitative interpretive case study methodology to examine the socio-spatial transformation of tourism in the coastal region of Indramayu, Indonesia. The empirical case was the Karangsong coastal area in Indramayu Regency. The research site of Karangsong was selected as it encompasses mangrove rehabilitation, coastal tourism, fishing and aquaculture activities, community livelihoods and tourism-related infrastructure within the same coastal landscape. Previous research has also identified Karangsong as an area with significant mangrove-based tourism potential and ongoing community and conservation activities. The study treats Karangsong as a socio-spatial case within the broader coastal context of Indramayu rather than as a representative sample of all coastal destinations in the regency. The empirical site encompasses the coastal and mangrove tourism area of Karangsong and the associated spaces in which the study participants experienced or observed tourism-related activities. These included tourism facilities and visitor areas, mangrove rehabilitation areas, fishing and livelihood spaces, tourism-related businesses, and community activity areas within the Karangsong coastal setting. This spatial boundary was maintained during interviews, observations, and documentary analysis so that the different forms of evidence referred to the same empirical case.

Fieldwork was conducted in September 2025 in the Karangsong coastal area. The sampling strategy using purposive sampling aimed to select participants who had direct experience, knowledge or involvement in tourism-related activities and socio-spatial changes in the Karangsong coastal area. The first author conducted all semi-structured interviews with 47 participants from six stakeholder groups: local residents, tourism operators and business owners, government and tourism officials, environmental or conservation actors, community leaders or organizations, and tourists or visitors. Potential participants were approached and invited to participate, with eligibility assessed according to their stakeholder role and ability to discuss tourism, livelihoods, environmental conditions, coastal change, or community relations. Interviews were conducted in Indonesian and generally lasted approximately 45–60 minutes. The first author also visited the study site during the fieldwork period to conduct

non-participant observations of tourism facilities and visitor areas, mangrove rehabilitation sites, fishing and livelihood spaces, tourism-related businesses, community activities, access routes, and interactions among different coastal uses. Documentary materials were selected purposively according to their relevance to the spatial, tourism, environmental, and development context of Karangsong. The documentary evidence provides further support for interpreting these accounts as part of an institutional arrangement in which tourism and conservation are intended to coexist. Regional Regulation No. 9 of 2024 (RTRW 2024–2044) (Indramayu Regency Government, 2024b) explicitly identifies Karangsong Beach and Mangrove Eco-tourism Karangsong as tourism locations. More importantly, Article 60 (Pasal 60) provides for mangrove conservation and reforestation involving communities, regular monitoring, and mangrove rehabilitation through a mangrove center for education and tourism. The same regulation also promotes mangrove ecotourism based on education and research and calls for the integration of marine tourism with spaces supporting conservation, aquaculture, and capture fisheries. These provisions contextualize the interview accounts by showing that the intended relationship between tourism and mangrove conservation is incorporated into the regional spatial-planning framework. However, the regulation establishes planning objectives rather than demonstrating actual ecological outcomes; therefore, it supports the institutional context of the findings but cannot by itself confirm whether conservation objectives have been achieved in practice. The triangulation process enabled the study to distinguish between participants' perceptions of change, contemporary conditions directly observed in the field, and formally stated regional planning objectives.

The qualitative data were analyzed using reflexive thematic analysis following Braun & Clarke's approach. Reflexive thematic analysis was selected because it enables researchers to interpret patterns of meaning across qualitative datasets while recognizing the active and reflexive role of the researcher in the analytical process (Braun & Clarke, 2019; Byrne, 2022). The analysis involved familiarization with the interview transcripts and field notes, generation of initial codes, development of candidate themes, review and refinement of themes, definition and naming of final themes, and interpretation of the findings in relation to the research questions and theoretical framework. Researcher reflexivity was considered throughout data collection and analysis. The first author's direct involvement in both interviewing and site observation provided contextual familiarity with the study area but also required attention to assumptions brought into the interpretation. The theory of the production of space (Lefebvre, 1991) initially sensitized the analysis to spatial practices, institutional representations, lived meanings, and stakeholder relations. During analysis, comparison of interview accounts with field observations and documentary evidence led to a more cautious interpretation of stakeholder differences. For example, differences between residents' livelihood concerns and tourism-development perspectives were initially considered as potential spatial conflict, but closer examination showed stronger evidence of the coexistence of different priorities than of a documented negotiation or direct conflict. The interpretation was therefore refined to describe overlapping stakeholder interests within a multifunctional coastal space rather than claiming conflict that was not directly demonstrated by the data. Reflexive notes and the audit trail were used to document these interpretive decisions. The three sources of evidence served different analytical purposes. Interviews were used to understand participants' recollections and interpretations of past and present coastal change. Observations documented spatial configurations and activities present during the September 2025 fieldwork. Planning documents provided evidence of formal development priorities and management objectives, rather than proof that proposed future outcomes had been achieved. Comparing these sources helped contextualize participants' accounts while maintaining the distinction between remembered change, observed conditions, and planned development.

Purposive sampling sought information-rich accounts from participants with substantive involvement in the case site, rather than statistical or demographic representativeness. As interviews progressed, the relevance of the accounts to the research questions and developing thematic analysis was considered. Recruitment ended when accounts across the six stakeholder groups were sufficiently diverse and information-rich, and additional participants no longer substantially expanded the range of perspectives represented. The decision considered both breadth across stakeholder groups and richness of the material.

To enhance trustworthiness, credibility was supported through triangulation across interviews, September 2025 non-participant observations, documentary materials, and comparison among the six stakeholder groups. Interviews were used to examine participants' experiences and interpretations, observations were used to contextualize contemporary spatial practices and physical conditions, and documentary materials were used to provide historical and institutional context. Triangulation was therefore used to identify convergence and qualification across evidence sources rather than to assume that each source independently verified the others.

All interviews were conducted in Indonesian. The interviews were transcribed in the original Indonesian language before coding so that the initial analysis remained as close as possible to participants' original expressions and meanings. Coding and theme development were conducted using the Indonesian-language transcripts, with English translations prepared only when participant quotations were selected for presentation in the manuscript. Selected quotations were translated into English while retaining the intended meaning, contextual nuance, and tone of the original statement. The translated quotations were compared with the corresponding Indonesian transcript to note any omission, change of meaning, or expressions that might be misleading in English. Where an Indonesian expression did not have a direct English equivalent, the translation prioritized conceptual meaning

rather than literal word-for-word correspondence. The final English quotations were reviewed against the original-language extracts before inclusion in the manuscript. Participant codes were retained consistently between the original transcripts and the English quotations. These procedures are consistent with contemporary qualitative approaches that emphasize methodological transparency, reflexivity, and coherence between research questions, data collection, and analytical procedures (Levitt et al., 2017). Ethical principles were applied throughout the research process. Participants were informed about the purpose of the study, the voluntary nature of their participation, confidentiality procedures, and their right to withdraw. Informed consent was obtained before interviews were conducted. Participant identities were protected through the use of codes rather than real names, such as R01 for residents, T01 for tourism operators, G01 for government officials, E01 for environmental actors, C01 for community representatives, and V01 for visitors. Interview recordings, transcripts, and field notes were securely stored and used solely for research purposes.

4. Results and Discussion

4.1 Overview of Participants

A total of 47 participants were selected using purposeful sampling wherein the criteria for selection were the competence of individuals, knowledge, and participation in coastal tourism development in Indramayu, as shown in Table 1.

Table 1. Overview of study participants

Stakeholder Group	Code	Number	Percentage	Primary Contribution to This Study
Local residents	R01–R15	15	31.9%	Everyday spatial practices, livelihoods, place identity
Tourism operators/business owners	T01–T09	9	19.1%	Tourism development, businesses, visitor activities
Government/tourism officials	G01–G06	6	12.8%	Planning, governance, infrastructure, tourism policy
Environmental/conservation actors	E01–E05	5	10.6%	Mangrove conservation, environmental change
Community leaders/organizations	C01–C05	5	10.6%	Community participation, local interests
Tourists/visitors	V01–V07	7	14.9%	Visitor experience, destination perception
Total		47		100.0%

Note: R = resident; T = tourism operator/business owner; G = government/tourism official; E = environmental/conservation actor; C = community leader/organization; V = visitor/tourist.

The participants were invited to the study from different stakeholder groups in order to have diverse perspectives on the socio-spatial changes discussed in the study, such as local residents, tourism entrepreneurs, government and tourism agencies representatives, environmentalists and conservationists, community leaders, and tourists. The local population contributed to the study with their knowledge about the local changes related to the daily practices, livelihood of people, accessibility to different places, and identity of the place whereas tourism businesses were included to analyze the impact of economic processes on the place; the representatives of the government and management were included in the study as they could have their opinion regarding planning and developing coastal space; the members of environmental groups took part in the research in order to express their opinion on the preservation of mangroves; community leaders were invited to represent their viewpoint on the involvement of the population; and tourists participated to share their experiences related to tourism in the changed location. Participant recruitment continued according to the relevance and richness of the information provided by participants, consistent with contemporary approaches to qualitative sample-size determination (Wutich et al., 2024).

4.2 Emerging Themes

As shown in Table 2, the five themes are analytically related to Lefebvre's spatial triad, but the dimensions are not treated as fixed categories assigned to particular objects or stakeholder groups. The same empirical phenomenon may involve more than one dimension depending on whether the analysis concerns its material use, institutional representation, or lived meaning. The table therefore identifies the primary analytical focus of each theme rather than suggesting a one-to-one correspondence between empirical objects and Lefebvre's spatial dimensions. During thematic refinement, the six preliminary analytical issues—changing spatial uses, access and control, livelihood opportunities, environmental conservation, place meanings, and future visions—were reviewed in relation to interviews, observations, and documents. Instead of being pre-determined themes, these issues worked as sensitizing concepts. The theme "Access, control and equity" was not kept as a separate theme due to

the fact that whenever the evidence concerning access to coastal area and tourism opportunities was presented, it was already part of the themes on spatial transformation and diversification of livelihoods. The preliminary issue "Visions for the future" is addressed cautiously within Theme 5 through stakeholder perspectives and priorities; the accounts presented do not establish specific future plans or decisions. Thus, five themes formed the basis of the investigation instead of six preliminary issues.

Table 2. Overview of themes and analytical interpretation

Theme	Main Codes	Lefebvre Dimensions	Analytical Focus
Transformation from livelihood space to multifunctional coastal space	Land use; tourism facilities; visitor activities; access	Perceived; conceived	Coexisting coastal uses, mobility, and planning representations
Tourism and livelihood diversification	Tourism jobs; small businesses; fishing; income opportunities	Perceived; lived	Livelihood diversification and participants' experiences of economic opportunities
Reconstruction of coastal place identity	Community pride; destination image; memories; local identity	Lived; conceived	Place meanings and attachments alongside destination representations
Coexistence of tourism and environmental conservation	Mangrove protection; environmental awareness; waste; ecological pressure	Perceived; conceived; lived	Relationships among tourism activities, conservation planning, and environmental perceptions
Stakeholder perspectives and priorities for coastal space	Stakeholder priorities; access; planning; participation; potential tensions	Perceived; conceived; lived	Stakeholder priorities across everyday uses, institutional planning, and lived experiences

Note: Perceived = spatial practices; conceived = representations of space; lived = representational spaces. These dimensions are interrelated rather than mutually exclusive.

4.2.1 Transformation from livelihood space to multifunctional coastal space

The first theme concerns the transformation of the coastal environment from a predominantly livelihood-oriented space into a multifunctional tourism landscape. Participants described the coast as historically associated with fishing, aquaculture, transportation, settlement, and everyday community activities. The September 2025 field observations provide a clearer picture of the contemporary spatial arrangement described by participants. The field notes documented tourism facilities and visitor areas alongside fishing and other livelihood spaces, mangrove rehabilitation areas, tourism-related businesses, and community activity areas. These observations indicate that tourism-related uses currently coexist with established livelihood and environmental functions. The increase and development of tourism have also led to the emergence of additional functions, such as recreational activities, visitor facilities, food businesses, commercial activities and tourism-related infrastructure. The historical shift from livelihood-oriented uses toward a multifunctional coastal landscape is therefore interpreted primarily through participants' recollections, contextualized by the available planning documents.

A local resident states: "*Before tourism became popular, people mainly came here for fishing and their daily activities. Now there are more visitors, places to eat, and tourism facilities, so the way we use the area has changed*." (R07)

A tourism operator notes: "*The coast is not only for local people anymore. Visitors come to take pictures, enjoy the mangrove, eat, and spend time with their families*." (T03)

The September 2025 observations documented tourism facilities, visitor activities, and food businesses alongside existing livelihood uses, supporting R07's and T03's descriptions of contemporary multifunctional use. R07 also recalls earlier fishing and community activities, whereas T03 emphasizes current recreational uses. These accounts reflect participants' recollections and perceptions; the observations cannot establish when changes began or verify historical conditions. The planning documents provide an institutional context for tourism's incorporation into coastal space. This suggests that tourism contributes to the reconfiguration of spatial practices. Space is produced through everyday practices, and changes in those practices can consequently transform the social character of a place (Duff, 2010). Tourism, therefore, does not merely occupy pre-existing coastal space; it also produces new spatial functions and relationships. This is in line with Green (2005) who demonstrated that tourism development can bring about profound changes in community space as tourism-related activities are embedded within existing social environments. Similarly, Terkenli (2025) asserted that tourism-related spatial transformation is shaped by interactions between economic interests, cultural meanings, and power relations. Taken together, the interview, observation, and documentary evidence support the interpretation that tourism has become an additional spatial function within an already multifunctional coastal area.

4.2.2 Tourism and livelihood diversification

The second theme concerns the relationship between tourism development and local livelihoods. Participants described tourism as creating additional economic opportunities through food businesses, parking services, guiding, transportation, accommodation, souvenir sales, and other small-scale activities. However, participants also indicated that tourism does not benefit all community members equally and does not necessarily replace traditional occupations. Field observations in September 2025 recorded food stalls, guiding activities, parking services and tourism-related businesses in the coastal tourism area.

Another local resident explains: "*Tourism gives us another source of income. Fishing is uncertain, but when visitors come, we can earn something from food and other small businesses*." (R12)

However, another participant highlights: "*Tourism helps some people more than others. Those who have businesses near the tourist area can earn more, while others still depend on fishing*." (R04)

These examples provide concrete evidence of the types of economic activities currently associated with tourism and contextualize R12's account of supplementary income opportunities. At the same time, R04's account indicates that access to these opportunities may differ among residents. The observations establish the presence of tourism-related activities, but they do not measure participants' income or demonstrate how economic benefits are distributed. In general, the evidence suggests that tourism can provide additional livelihood opportunities, while traditional fishing activities continue to be part of the local economy. This finding is consistent with Quang et al. (2023) who found that tourism development in a coastal community could create alternative livelihood opportunities but also challenges of participation and unequal involvement. From the perspective of tourism geography this is significant as the question should not be whether tourism produces economic benefits but who has access to those benefits and how the transformation of space affects existing livelihood practices (Gibson, 2009). The findings therefore suggest that tourism planning in Indramayu should consider livelihood continuity and equitable economic participation alongside visitor growth.

4.2.3 Reconstruction of coastal place identity

The third theme concerns changes in the symbolic and emotional meaning of the coastal environment. Participants did not describe the coast exclusively in economic or physical terms. Some accounts indicated that tourism had contributed to greater recognition of the area and strengthened community pride in the mangrove environment and local destination.

A local resident states: "*People from outside Indramayu now know this place because of the mangrove and tourism. It makes us proud because the area has become known as a destination*." (R09)

Similarly, a community participant explains: "*The mangrove used to be seen mainly as something that needed to be protected. Now people also see it as part of the identity of the tourism destination.*" (C03)

These narratives illustrate the transformation of the coast from an environmental and livelihood resource into a symbolically meaningful tourism place. This relates particularly to representational spaces or lived space, where physical environments acquire meaning through memory, experience, identity, and symbolism. This finding extends previous research on Karangsong, which has identified the area's mangrove resources and ecotourism potential (Garsetiasih et al., 2021; Novianti et al., 2022; Prihadi et al., 2024). While these studies establish the ecological and tourism value of the area, the present study focuses on how tourism contributes to changing the meaning of the place itself. Tourism can therefore be understood as participating in a process of place-making. The mangrove is no longer understood solely as an ecological resource; it becomes simultaneously an attraction, a destination symbol, an educational environment, and a source of community identity.

4.2.4 Coexistence of tourism and environmental conservation

The fourth theme deals with the complex relationship between tourism and environmental conservation. Most participants acknowledged the potential of tourism to increase awareness of the importance of mangroves and to create economic incentives for their conservation. However, concerns were also expressed regarding waste, visitor pressure, infrastructure development and possible environmental degradation.

An environmental stakeholder explains, "*Tourism makes people more aware that mangroves are important, but if the number of visitors increases without proper management, there can also be waste and damage*." (E02)

A community participant explains, "*Now people understand that protecting the mangrove is important because it protects the coast and also brings visitors and income*." (C02)

These accounts illustrate different perceptions of the relationship between tourism and mangrove conservation in Karangsong. Environmental participant E02 expresses a conditional concern that increasing visitor numbers could generate waste and damage if tourism is not properly managed. This statement identifies a perceived environmental risk. In contrast, community participant C02 emphasizes the perceived value of mangrove protection for coastal protection, tourism, and local income. This account illustrates the perceived compatibility of conservation and livelihood benefits, but it does not independently demonstrate measurable improvements in ecological conditions or conservation effectiveness. Tourism can increase the visibility and economic value of environmental resources, but tourism itself can also become a source of environmental pressure. The result that tourism and conservation are not necessarily mutually reinforcing is in line with studies that describe the complex

relations between economic development, environmental vulnerability and spatial adaptation within the framework of coastal tourism (Jarratt & Davies, 2020). From a socio-spatial perspective, the conversion of the mangrove into a tourist attraction changes its social representation (Nurhasanah & Van den Broeck, 2022; Oldenburg, 2025).

The documentary evidence provides an institutional context for these accounts. Regional Regulation No. 9 of 2024 on the Indramayu Regency Spatial Plan (RTRW 2024–2044) (Indramayu Regency Government, 2024b) identifies Karangsong Beach and Mangrove Eco-Tourism Karangsong as tourism locations. Article 60 (Pasal 60) addresses mangrove conservation and reforestation, community participation, monitoring, and a mangrove center for education and tourism. The regulation also promotes mangrove ecotourism based on education and research and the integration of marine tourism with spaces supporting conservation, aquaculture, and capture fisheries. These provisions show that tourism development and mangrove conservation are both recognized within the regional planning framework. They describe planning objectives rather than verified ecological outcomes, so they contextualize participants' accounts without establishing whether environmental conditions have improved or deteriorated in practice. Regional Regulation No. 8 of 2024 on the Long-Term Regional Development Plan (RPJPD 2025–2045) (Indramayu Regency Government, 2024a) provides the broader regional development context. The mangrove becomes simultaneously a conservation space and tourism commodity. The challenge for sustainable tourism governance is therefore to prevent its economic representation as an attraction from undermining its ecological and community functions.

4.2.5 Stakeholder perspectives and priorities for coastal space

The fifth theme concerns stakeholders' perspectives and priorities regarding the use and development of Karangsong's coastal space. The interviews indicate different priorities concerning livelihood continuity, tourism development, environmental protection, community benefits, and visitor experience. These accounts are interpreted as expressions of stakeholder priorities rather than as evidence of specific future plans or negotiated decisions. These findings show that Karangsong is a multifunctional coastal space shaped by different stakeholder interests. Residents emphasize livelihood continuity, tourism operators emphasize economic opportunities, government actors emphasize development and planning, environmental actors prioritize mangrove conservation, and visitors seek recreational experiences. Thus, tourism, fishing, and conservation do not operate in separate spaces but increasingly overlap within the same coastal landscape. In a Lefebvrian sense, this co-habitation is an example of how the three moments of the production of space (spatial practices, institutional representations and lived meanings) are articulated in the production and reconfiguration of coastal space.

A local resident explains: "*For local people, this is our livelihood. For tourists, it is a place to visit. At times what tourists want is different than what local people need.*" (R03)

A government representative notes: "*We want tourism to grow because it can benefit the community, but development also has to consider the environment and local needs*." (G02)

Meanwhile, a tourist describes the area from an experiential perspective: "*I came because I wanted to see the mangrove and experience a different kind of tourism near the coast*." (V04)

The accounts illustrate different priorities for shared coastal space. R03 emphasizes livelihood continuity and differences between residents' needs and tourists' preferences; G02 highlights environmental protection and community needs alongside tourism development; and V04 describes a recreational and experiential destination. These perspectives suggest potentially competing interests, but do not establish the extent of overlapping spatial uses or document specific interactions, decisions, compromises, or disputes. They are therefore interpreted as stakeholder priorities and potential tensions rather than evidence of completed negotiation.

From the perspective of Lefebvre (1991) Production of Space, these findings demonstrate the coexistence and interaction of spatial practices, representations of space, and lived experiences. Residents' everyday livelihood practices coexist with government planning, tourism-oriented activities, conservation practices, and visitors' experiences. Similar studies have indicated that the spatial transformation related to tourism is the outcome of the interactions among economic interests, cultural meanings, institutional arrangements, and power relations (Kulusjärvi, 2020; Sun et al., 2020). Yang et al. (2026) likewise demonstrate that tourism transformation can involve the coexistence of tourism-oriented spatial uses and residents' continued lived-space practices. The evidence from this theme supports the interpretation of coastal tourism as a socio-spatial transformation involving multiple stakeholder meanings and interests.

4.3 Integrative Discussion: Coastal Tourism as Socio-Spatial Transformation

The five themes show that coastal tourism in Karangsong, Indramayu, involves spatial, economic, environmental, symbolic, and social transformations. Tourism introduces new spatial practices and economic opportunities, contributes to the reconstruction of place identity, creates new representations of environmental resources, and generates new relationships among stakeholders. The findings therefore extend the empirical themes into a broader socio-spatial interpretation of coastal tourism development. First, tourism modifies existing

spatial practices through the introduction of visitor facilities, commercial activities, recreational uses and new patterns of accessibility into spaces previously associated primarily with fishing, aquaculture, settlement and community activities. Secondly, these spatial changes are accompanied by livelihood diversification, as tourism generates complementary income opportunities, with traditional occupations continuing. Third, tourism contributes to place-making by changing how the coastal environment, particularly the mangrove landscape, is represented and experienced. Fourth, tourism creates both opportunities and pressures for environmental conservation, making the mangrove simultaneously an ecological resource, conservation space, and tourism attraction. Finally, these processes bring different stakeholder interests into the same geographical space, requiring continuing interaction over access, development priorities, conservation, and community benefits. The Karangsong case extends the application of Lefebvre's (1991) Production of Space by showing that coastal tourism does not necessarily replace existing livelihood and conservation spaces. Instead, tourism, fishing, aquaculture, mangrove conservation, community activities, and recreation become layered within the same coastal landscape. The transformation is therefore characterized not by a simple shift from "traditional space" to "tourism space," but by the reconfiguration and coexistence of multiple spatial functions and meanings. This process is shaped by residents' livelihood interests, tourism actors' economic interests, government planning priorities, conservation objectives, and visitors' recreational demands. The case consequently highlights multifunctionality and overlapping stakeholder interests as important mechanisms through which tourism space is socially produced.

Findings also suggest potential tensions between these different functions and interests. Tourism can offer livelihood opportunities and raise the visibility of mangrove conservation, but also raises issues around access, environmental pressure, benefit sharing and changing use of coastal resources. These tensions echo earlier tourism research that has suggested that spatial transformation includes relations between economic interests, cultural meanings, institutions, communities and power relations (Buzinde & Manuel-Navarrete, 2013; Kulusjärvi, 2020; Sun et al., 2020). Overall, the broader evidence presented in this study primarily demonstrates the coexistence of different meanings, interests, and potential tensions. Accordingly, the contribution of the study is more cautiously framed as a socio-spatial perspective that explains how tourism transforms coastal space while bringing different stakeholder priorities into the same landscape. This extends previous research on coastal tourism by Britton (1991) and Saarinen (2004) showing how the transformation of tourism destinations involves not only economic and environmental changes but also changes in the social production and lived meaning of space. Environmental gains are often incremental and require complementary organizational/government/institution strategies to achieve substantial impact (Jamalulel et al., 2026). The stakeholder accounts therefore point to different priorities and potential tensions rather than documented negotiations. The data do not demonstrate that these differences have resulted in formal bargaining or agreed decisions, nor do they establish how future development choices will be resolved. From these perspectives, sustainable coastal tourism requires attention to the multifunctionality of coastal space and to the interests of the communities, tourism actors, government institutions, environmental groups, and visitors who use and experience it. Community participation, livelihood continuity, environmental protection, equitable access to tourism opportunities, and spatial governance should therefore be considered interconnected dimensions of coastal transformation. The findings suggest that tourism planning should incorporate local knowledge and everyday practices when considering infrastructure, access, environmental management, visitor capacity, and future tourism development. The RPJPD 2025–2045 provides a broader regional development context (Indramayu Regency Government, 2024a). More specifically, the RTRW 2024–2044 provisions discussed above frame the planned relationship among tourism, mangrove conservation, aquaculture, and fisheries (Indramayu Regency Government, 2024b). Taken together, the interview, observation, and documentary evidence indicate that Karangsong's socio-spatial transformation involves the coexistence of tourism, livelihoods, and conservation within a shared coastal landscape, while the timing of historical changes and the achievement of future planning objectives remain beyond what the available evidence can conclusively establish.

5. Conclusion

This study analyzed the socio-spatial transformation of coastal tourism in Karangsong, Indramayu, Indonesia, using an interpretive qualitative approach with 47 participants representing local residents, tourism operators, government officials, environmental and conservation actors, community organizations, and tourists. By including multiple stakeholder perspectives, the study conceptualizes coastal tourism as not only the development of a tourism destination, but also as a process of socio-spatial transformation in which coastal space serves multiple purposes, acquires different meanings, and reflects diverse stakeholder priorities concerning tourism development, local livelihoods, and environmental conservation. The findings demonstrate that tourism development could transform the function and meaning of coastal space. Coastal areas that were previously associated with fishing, aquaculture, settlement, and everyday community activities are increasingly used for tourism, recreation, commercial activities, and environmental education. However, the transformation does not necessarily imply a total substitution of traditional spaces by tourism spaces. Instead, tourism-oriented activities and existing livelihood practices are found to coexist in the same geographical landscape.

The study additionally emphasizes the ambiguous link between tourism development and environmental conservation. Tourism can increase the visibility and economic value of mangroves and coastal ecosystems, and can potentially generate community support for conservation. But increased visitor activity and tourism infrastructure may lead to further environmental pressures. Therefore, tourism and conservation should not be assumed to be mutually reinforcing by default. Their relationship is determined by the success of spatial planning, environmental governance, community participation and distribution of tourism benefits. Understanding these processes is important for the development of coastal tourism that is not only economically viable but also socially inclusive, environmentally responsible and sensitive to the lived meanings of place. The study contributes to Lefebvre's Production of Space perspective by demonstrating how coastal tourism can be produced through the layering and reconfiguration of existing livelihood, conservation, and community functions rather than through their complete replacement. In Karangsong, the same coastal space is simultaneously a livelihood environment, tourism destination, conservation area, and lived community landscape. Its transformation therefore reflects the interaction of different stakeholder interests, institutional priorities, everyday practices, and place meanings.

This study has several limitations. First, participants' accounts of past coastal changes may be affected by retrospective recall and should be interpreted alongside documentary evidence rather than treated as direct observations of historical conditions. Second, purposive recruitment and access to available participants across stakeholder groups may have limited the range of perspectives represented. Third, the fieldwork was performed during a single period in September 2025, so the observations reflect the conditions at that time and cannot independently establish seasonal variation or long-term changes. Limited detail about recruitment routes, individual observation visits, and the researchers' prior relationships with the site also constrains assessment of how access and researcher positioning may have shaped the perspectives represented and their interpretation. The findings offer a contextual reading of socio-spatial transformation in Karangsong, rather than a full account of the experiences of all stakeholders or of coastal destinations in Indramayu. Future research could extend the use of repeated field visits, longer-term observation and broader stakeholder engagement to explore how spatial practices, livelihood opportunities and conservation priorities change over time.

Author Contributions

Conceptualization, I.J.; methodology, I.J.; formal analysis, I.J.; investigation, I.J.; resources, I.J.; data curation, I.J. and L.-W.L.; writing—original draft preparation, I.J. and L.-W.L.; writing—review and editing, L.-W.L.; visualization, I.J.; supervision, L.-W.L.; project administration, L.-W.L.; funding acquisition, L.-W.L. All authors have read and agreed to the published version of the manuscript.

Informed Consent Statement

Participation was voluntary, and informed consent was obtained from all participants. Participants were assured of confidentiality, and identifying information was removed before analysis.

Data Availability

The data supporting the findings of this study are available from the corresponding author upon request. The data are not publicly available due to participant privacy and confidentiality considerations.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

Generative AI and AI-assisted technologies were used to assist with language editing, improve readability, and support the design and preparation of selected figures. The authors take full responsibility for the accuracy, originality, and integrity of the manuscript, including all figures.

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